SUBOXONE® FILM
2MG/0.5MG & 8MG/2MG

NAME OF THE MEDICINE
SUBOXONE FILM contains buprenorphine (as hydrochloride) and naloxone (as hydrochloride dihydrate) at a ratio of 4:1 buprenorphine: naloxone (ratio of free bases).

DESCRIPTION
SUBOXONE FILM is a soluble film intended for sublingual and/or buccal administration only.

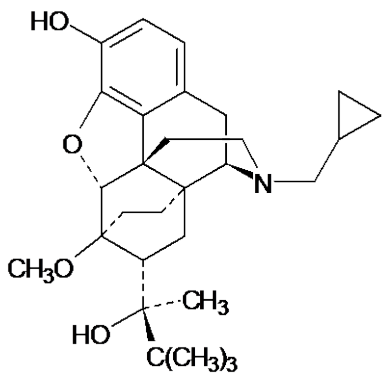
Each SUBOXONE FILM 2MG/0.5MG contains 2 mg buprenorphine (as hydrochloride) and 0.5 mg naloxone (as hydrochloride dihydrate). Each SUBOXONE FILM 2MG/0.5MG is an orange rectangular soluble film imprinted with “N2” in white ink.

Each SUBOXONE FILM 8MG/2MG contains 8 mg buprenorphine (as hydrochloride) and 2 mg naloxone (as hydrochloride dihydrate). Each SUBOXONE FILM 8MG/2MG is an orange rectangular soluble film imprinted with “N8” in white ink.

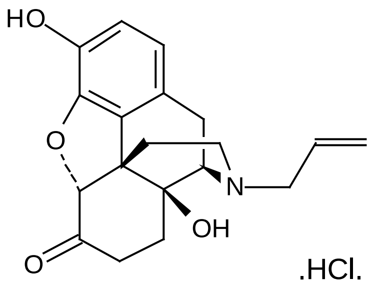
Inactive Ingredients: acesulfame potassium, citric acid, maltitol solution, hypromellose, polyethylene oxide, sodium citrate, lime flavour, Sunset Yellow FCF and a white printing ink.

Buprenorphine hydrochloride is a white powder, weakly acidic with limited solubility in water (19.5 mg /mL at 37°C, pH 4.1). Chemically, it is 21- Cyclopropyl-7α-[(*S*) -1- hydroxy-1, 2, 2 - trimethylpropyl]-6, 14-endo-ethano-6, 7, 8, 14-tetrahydrooripavine hydrochloride. Buprenorphine hydrochloride has the molecular formula $C_{29}H_{41}NO_4$ HCl and the molecular weight is 504.09. The CAS number is 53152-21-9.

Naloxone hydrochloride is a white to slightly off-white powder that exists as the dihydrate and is soluble in water, in dilute acids and in strong alkali. Chemically, it is (-)-17-Allyl-4, 5α-epoxy-3, 14-dihydroxymorphinan-6-one hydrochloride dihydrate. Naloxone hydrochloride has the molecular formula $C_{19}H_{21}NO_4$ HCl .$2H_2O$ and the molecular weight is 399.87. The CAS number of naloxone hydrochloride dihydrate is 51481-60-8. The chemical structures of buprenorphine hydrochloride and naloxone hydrochloride dihydrate are:


.HCl

Buprenorphine hydrochloride


.HCl. $2H_2O$

Naloxone hydrochloride dihydrate

PHARMACOLOGY
Pharmacodynamic properties
Buprenorphine is a µ (mu) opioid receptor partial agonist, κ (kappa) opioid receptor antagonist. Its activity in opioid maintenance treatment is attributed to its slow dissociation from the µ receptors in the brain which reduces craving for opioids and opioid withdrawal symptoms. This minimises the need of the opioid dependent patient for illicit opioid medicines.

During clinical pharmacology studies in opioid-dependent subjects, buprenorphine demonstrated a ceiling effect on a number of parameters, including positive mood, “good effect” and respiratory depression.

Naloxone is an antagonist at µ (mu), δ (delta) and κ (kappa) opioid receptors. Because of its almost complete first pass metabolism, naloxone administered orally, sublingually or buccally has no detectable pharmacological activity. However, when administered intravenously to opioid dependent persons, the presence of naloxone in SUBOXONE FILM produces marked opioid antagonist effects and opioid withdrawal, thereby deterring intravenous abuse.

PRECLINICAL TOXICOLOGY

The combination of buprenorphine and naloxone has been investigated in acute and repeated dose (up to 90 days in rats) toxicity studies in animals. No synergistic enhancement of toxicity has been observed. Undesirable effects were based on the known pharmacological activity of opioid agonistic and/or antagonistic substances.

The combination (4:1) of buprenorphine hydrochloride and naloxone hydrochloride was not mutagenic in a bacterial mutation assay (Ames test) and was not clastogenic in an in vitro cytogenetic assay in human lymphocytes or in an intravenous micronucleus test in the rat.

Effects on fertility

Dietary administration of buprenorphine/naloxone in the rat at dose levels of 500 ppm or greater produced a reduction in fertility demonstrated by reduced female conception rates.

A dietary dose of 100 ppm (estimated exposure approximately 2.4x for buprenorphine at a human dose of 24 mg buprenorphine/naloxone based on AUC, plasma levels of naloxone were below the limit of detection in rats) had no adverse effect on fertility in females.

Carcinogenicity

A carcinogenicity study with buprenorphine/naloxone was conducted in rats at doses of 7, 31 and 123 mg/kg/day, with estimated exposure multiples of 4, 18 and 44 times, based on a human daily sublingual dose of 16 mg calculated on a mg/m^2 basis. Statistically significant increases in the incidence of benign testicular interstitial (Leydig's) cell adenomas were observed in all dosage groups.

Pharmacokinetic properties

Buprenorphine:

<u>Absorption</u>

When taken orally, buprenorphine undergoes first-pass metabolism with N-dealkylation and glucuronidation in the small intestine and the liver. The use of SUBOXONE FILM by the oral route is therefore inappropriate. SUBOXONE FILMS are for sublingual and/or buccal administration.

Table 1 shows the pharmacokinetic parameters of buprenorphine, norbuprenorphine, and naloxone after administration of SUBOXONE FILM in randomised, crossover studies. Overall, there was wide variability in the sublingual absorption of buprenorphine and naloxone. SUBOXONE FILM and SUBOXONE sublingual tablet do not meet all criteria for bioequivalence. Patients being switched between tablets and soluble films may therefore require dosage adjustment (see **DOSAGE AND ADMINISTRATION**).

In several pharmacokinetic studies following the administration of different dosages, a dose of one or two of the 2 mg/0.5 mg SUBOXONE FILMS administered sublingually or buccally showed comparable relative bioavailability to the same total dose of SUBOXONE sublingual tablets. In contrast, one 8 mg/2 mg and one 12 mg/3 mg SUBOXONE FILM administered sublingually or buccally showed higher relative bioavailability for both buprenorphine and naloxone compared to the same total dose of SUBOXONE sublingual tablets. A combination of one 8 mg/2 mg and two 2 mg/0.5 mg SUBOXONE FILMS (total dose of 12 mg/ 3 mg) administered sublingually showed comparable relative bioavailability to the same total dose of SUBOXONE sublingual tablets, while buccally administered SUBOXONE FILMS showed higher relative bioavailability. Table 2, below, illustrates the relative increase in exposure to buprenorphine and naloxone associated with SUBOXONE FILMS compared to SUBOXONE sublingual tablets, and shows the effect of route of administration.

Table 1. Pharmacokinetic parameters (Mean ± SD) of buprenorphine and naloxone following SUBOXONE FILM administration

PK Parameter	SUBOXONE Film Dose (mg)			
	2 mg/0.5 mg	**4 mg / 1 mg**	**8 mg / 2 mg**	**12 mg / 3 mg**
Buprenorphine				
C_{max} (ng/mL)	0.947 ± 0.374	1.40 ± 0.687	3.37 ± 1.80	4.55 ± 2.50
T_{max} (h) Median, (min-max)	1.53 (0.75 - 4.0)	1.50 (0.5, 3.0)	1.25 (0.75 - 4.0)	1.50 (0.5, 3.0)
AUC_{inf} (ng.hr/mL)	8.654 ± 2.854	13.71 ± 5.875	30.45 ± 13.03	42.06 ± 14.64
$t_{1/2}$ (hr)	33.41 ± 13.01	24.30 ± 11.03	32.82 ± 9.81	34.66 ± 9.16
Norbuprenorphine				
C_{max} (ng/mL)	0.312 ±0.140	0.617 ±0.311	1.40 ±1.08	2.37 ±1.87
T_{max} (h) Median, (min-max)	1.38 (0.5 - 8.0)	1.25 (0.5, 48.0)	1.25 (0.75 - 12.0)	1.25 (0.75, 8.0)
AUC_{inf} (ng.hr/mL)	14.52 ±5.776	23.73 ±10.60	54.91 ±36.01	71.77 ±29.38
$t_{1/2}$ (hr)	56.09 ±31.14	45.96 ±40.13	41.96 ±17.92	34.36 ±7.92
Naloxone				
C_{max} (ng/mL)	0.054 ± 0.023	0.0698 ± 0.0378	0.193 ± 0.091	0.238 ± 0.144
T_{max} (h) Median, (min-max)	0.75 (0.5 - 2.0)	0.75 (0.5, 1.5)	0.75 (0.5 - 1.25)	0.75 (0.50, 1.25)
AUC_{inf} (ng.hr/mL)	0.137 ± 0.043	0.204 ± 0.108	0.481 ± 0.201	0.653 ± 0.309
$t_{1/2}$ (hr)	5.00 ± 5.52	3.91 ± 3.37	6.25 ± 3.14	11.91 ± 13.80

Table 2. Changes in Pharmacokinetic Parameters for SUBOXONE FILM Administered Sublingually or Buccally in Comparison to SUBOXONE sublingual tablet								
Dosage	PK Parameter	**Increase in Buprenorphine**			PK Parameter	**Increase in Naloxone**		
		Film Sublingual Compared to Tablet Sublingual	Film Buccal Compared to Tablet Sublingual	Film Buccal Compared to Film Sublingual		Film Sublingual Compared to Tablet Sublingual	Film Buccal Compared to Tablet Sublingual	Film Buccal Compared to Film Sublingual
1 x 2 mg/0.5 mg	C_{max}	22%	25%	-	C_{max}	-	-	-
	$AUC_{0\text{-}last}$	-	19%	-	$AUC_{0\text{-}last}$	-	-	-
2 x 2 mg/0.5 mg	C_{max}	-	21%	21%	C_{max}	-	17%	21%
	$AUC_{0\text{-}last}$	-	23%	16%	$AUC_{0\text{-}last}$	-	22%	24%
1 x 8 mg/2 mg	C_{max}	28%	34%	-	C_{max}	41%	54%	-
	$AUC_{0\text{-}last}$	20%	25%	-	$AUC_{0\text{-}last}$	30%	43%	-
1 x 12 mg/3 mg	C_{max}	37%	47%	-	C_{max}	57%	72%	9%
	$AUC_{0\text{-}last}$	21%	29%	-	$AUC_{0\text{-}last}$	45%	57%	-
1 x 8 mg/2 mg plus 2 x 2 mg/0.5 mg	C_{max}	-	27%	13%	C_{max}	17%	38%	19%
	$AUC_{0\text{-}last}$	-	23%	-	$AUC_{0\text{-}last}$	-	30%	19%

Note: 1. '–' represents no change when the 90% confidence intervals for the geometric mean ratios of the C_{max} and $AUC_{0\text{-}last}$ values are within the 80% to 125% limit. 2. There is no data for the 4 mg/1 mg strength film; it is compositionally proportional to 2 mg/0.5 mg strength film and has the same size of 2 x 2 mg/0.5 mg film strength.

Distribution

The absorption of buprenorphine is followed by a rapid distribution phase (distribution half-life of 2 to 5 hours). Buprenorphine is highly lipophilic which leads to rapid penetration of the blood-brain barrier. The medicine is around 96% protein bound primarily to alpha and beta globulin.

Metabolism and elimination

In animals and man buprenorphine is metabolised by Phase 1 (oxidative) and Phase 2 (conjugation) reactions. It is oxidatively metabolised by N-dealkylation and glucuroconjugation of the parent molecule and the dealkylated metabolite. Clinical data confirm that CYP3A4 is responsible for the N-dealkylation of buprenorphine. N-dealkylbuprenorphine is a µ (mu)-opioid agonist with weak intrinsic activity.

Elimination of buprenorphine is bi- or tri-exponential, with a long terminal elimination phase (refer to Table 1), due in part to re-absorption of buprenorphine after intestinal hydrolysis of the conjugated metabolite, and in part to the highly lipophilic nature of the molecule. Buprenorphine is essentially eliminated in the faeces by biliary excretion of the glucuroconjugated metabolites (70%), the rest being eliminated in the urine.

Naloxone:

Absorption and distribution

Following intravenous administration, naloxone is rapidly distributed (distribution half-life of around 4 minutes). Naloxone is approximately 45% protein bound, primarily to albumin. Following oral administration, naloxone is barely detectable in plasma; following sublingual administration of SUBOXONE FILM, plasma naloxone concentrations are low and decline rapidly.

Metabolism and elimination

Naloxone is metabolised in the liver, primarily by glucuronide conjugation, and excreted in the urine. Naloxone has a short elimination half-life (refer to Table 1).

Special Populations

Elderly

No pharmacokinetic data in elderly patients are available.

Renal impairment

Renal elimination plays a relatively small role (~30%) in the overall clearance of SUBOXONE FILM. No dose modification based on renal function is generally required. Metabolites of buprenorphine accumulate in patients with renal failure. Caution is recommended when dosing subjects with severe renal impairment, which may require dose adjustment.

Hepatic impairment

Hepatic elimination plays a relatively large role (~70%) in the overall clearance of SUBOXONE FILM and the action of buprenorphine may be prolonged in subjects with impaired hepatic clearance. Lower initial SUBOXONE FILM doses and cautious titration of dosage may be required in patients with mild to moderate hepatic dysfunction (see **PRECAUTIONS**).

SUBOXONE FILM is contraindicated in patients with severe hepatic dysfunction (see **CONTRAINDICATIONS**).

Clinical Efficacy

Efficacy and safety data for SUBOXONE sublingual tablets are primarily derived from a one-year clinical trial, comprising a 4-week randomised double blind comparison of SUBOXONE sublingual tablets, buprenorphine and placebo tablets followed by a 48 week safety study of SUBOXONE sublingual tablets. In this trial, 326 heroin-dependent subjects were randomly assigned to either SUBOXONE sublingual tablets 16 mg per day, 16 mg buprenorphine per day or placebo tablets.

For subjects randomised to either active treatment, dosing began with one 8 mg tablet of buprenorphine on Day 1, followed by 16 mg (two 8 mg tablets) of buprenorphine on Day 2. On Day 3, those randomised to receive SUBOXONE sublingual tablet were switched to the combination tablet. Subjects were seen daily in the clinic (Monday through Friday) for dosing and efficacy assessments. Take-home doses were provided for weekends. The primary study comparison was to assess the efficacy of buprenorphine and SUBOXONE sublingual tablet individually against placebo. The percentage of thrice-weekly urine samples that were negative for non-study opioids was statistically higher for both SUBOXONE sublingual tablet versus placebo ($p < 0.0001$) and buprenorphine versus placebo ($p < 0.0001$).

In a double-blind, double-dummy, parallel-group study comparing buprenorphine ethanolic solution to a full agonist active control, 162 subjects were randomised to receive the ethanolic sublingual solution of buprenorphine at 8 mg/day (a dose which is roughly comparable to a dose of 12 mg/day of SUBOXONE sublingual tablets), or two relatively low doses of active control, one of which was low enough to serve as an alternative to placebo, during a 3 to 10 day induction phase, a 16-week maintenance phase and a 7-week detoxification phase. Buprenorphine was titrated to maintenance dose by Day 3; active control doses were titrated more gradually. Based on retention in treatment and the percentage of thrice-weekly urine samples negative for nonstudy opioids, buprenorphine was more effective than the low dose of the control, in keeping heroin addicts in treatment and in reducing their use of opioids while in treatment. The effectiveness of buprenorphine, 8 mg per day was similar to that of the moderate active control dose, but equivalence was not demonstrated.

In an open-label study, SUBOXONE FILM was administered sublingually for 12 weeks. A total of 194 participants were enrolled in the study and 118 participants received and completed 12 weeks of SUBOXONE FILM treatment. Treatment was conducted on an outpatient basis and included study drug dosing as well as safety and tolerability assessments.

The sublingual films were initially given in doses identical to the participant's established daily maintenance dose of 4 mg/1 mg to 32 mg/8 mg of buprenorphine/naloxone. Dose adjustments within this range were allowed at the discretion of the Investigator. Throughout this study, the number of participants requiring a dose adjustment changed with each treatment visit. Depending on the visit, 76.6%-99.4% of participants required no dose adjustments to their SUBOXONE FILM daily dose. Most dose changes occurred after the first 4 weeks of the treatment period, indicating that the transfer from the SUBOXONE sublingual tablet to the SUBOXONE FILM was adequate and the difference between the formulations was not likely to be clinically significant. These were changes that may normally be observed through the course of treatment. Due to the potential individual variability in the bioavailability of SUBOXONE FILM compared with SUBOXONE sublingual tablet, patients switching from the film to the tablet should be monitored for potential dose decrease or increase. The safety results from this study showed that the SUBOXONE FILM is safe and well tolerated.

INDICATIONS

Substitution treatment for opioid drug dependence, within a framework of medical, social and psychological treatment. The intention of the naloxone component is to deter intravenous misuse. Treatment is intended for use in adults and adolescents over 15 years of age who have agreed to be treated for addiction.

DOSAGE AND ADMINISTRATION

The routes of administration of SUBOXONE FILM are sublingual and buccal only. The film formulation is not designed to be split or broken.

Treatment must be under the supervision of a physician experienced in the management of opioid dependence/addiction.

Each SUBOXONE FILM contains buprenorphine and naloxone.
It should not be swallowed whole as this reduces the bioavailability of the medicine. Physicians must warn patients that the sublingual and buccal route are the only effective and safe routes of administration for this medicine (see **PRECAUTIONS**).

Please note: The following instructions refer to the buprenorphine content of each dose.

Method of Administration

Sublingual Administration

Place one film under the tongue, close to the base on the left or right side. If an additional film is necessary to achieve the prescribed dose, place an additional film sublingually on the opposite side from the first film. Place the film in a manner to minimize overlapping as much as possible. The film must be kept under the tongue until the film is completely dissolved. If a third film is necessary to achieve the prescribed dose, place it under the tongue on either side after the first 2 films have dissolved.

Buccal Administration

Place one film on the inside of the right or left cheek. If an additional film is necessary to achieve the prescribed dose, place an additional film on the inside of the opposite cheek. The film must be kept on the inside of the cheek until the film is completely dissolved. If a third film is necessary to achieve the prescribed dose, place it on the inside of the right or left cheek after the first two films have dissolved.

No food or drink should be consumed until the film is completely dissolved. SUBOXONE FILM should NOT be chewed, swallowed, or moved from placement.

Adults
Baseline liver function tests and documentation of viral hepatitis status are recommended prior to commencing therapy. Patients who are positive for viral hepatitis, on concomitant medicines (see **DRUG INTERACTIONS**) and/or have existing liver dysfunction are at greater risk of liver injury. Regular monitoring of liver function is recommended (see **PRECAUTIONS**).

<u>Induction</u>
An adequate maintenance dose, titrated to clinical effectiveness, should be achieved as rapidly as possible to prevent undue opioid withdrawal symptoms due to inadequate dosage.

Prior to induction, consideration should be given to the type of opioid dependence (i.e. long- or short-acting opioid), the time since last opioid use and the degree or level of opioid dependence. To avoid precipitating withdrawal, induction with SUBOXONE FILM should be undertaken when objective and clear signs of withdrawal are evident.

Due to naloxone exposure being somewhat higher following buccal administration than sublingual administration, it is recommended that the sublingual site of administration be used during induction to minimise naloxone exposure and to reduce the risk of precipitated withdrawal.

When initiating buprenorphine treatment, the physician should be aware of the partial agonist profile of the molecule to the μ opioid receptors, which can precipitate withdrawal in opioid-dependent patients if given too soon after the administration of heroin, methadone or another opioid.

<u>Patients taking Heroin (or Other Short-acting Opioids)</u>
When treatment starts, the first dose of SUBOXONE FILM should be taken at least 6 hours after the patient last used opioids and when the objective signs of withdrawal appear.

The Clinical Opiate Withdrawal Scale (COWS) may be a useful reference assessment however clinical assessment of withdrawal symptoms with consideration of the patient’s baseline presentation is important, particularly for patients in mild withdrawal (COWS score of 5-12). The recommended starting dose is 4-8 mg SUBOXONE FILM on Day One, with a possible additional 4 mg depending on the individual patient’s requirement. The suggested target total dose for Day One is in the range of 8-12 mg SUBOXONE FILM. For patients with moderate or severe withdrawal at the time of the first dose, an initial dose of 8 mg may be appropriate with an additional 4 mg depending on the individual patient’s requirement to a total maximum of 12 mg on Day 1.

Lower doses (e.g. 2 or 4 mg total on Day 1) are suited to those with low or uncertain levels of opioid dependence, with high risk polydrug use (alcohol, benzodiazepines) or with other severe medical complications. Seek specialist advice if concerned.

<u>Patients on Methadone</u>
Before starting treatment with SUBOXONE FILM, the maintenance dose of methadone should be reduced to the minimum methadone daily dose that the patient can tolerate. The first dose of SUBOXONE FILM should be taken at least 24 hours after the patient last used methadone.

An initial dose of 2 mg SUBOXONE FILM may be administered when moderate withdrawal is apparent (COWS ≥ 13). An additional dose of 6 mg SUBOXONE FILM can be administered one hour later if the initial dose does not precipitate withdrawal. Supplementary doses can be administered every 1 to 3 hours according to withdrawal severity:

- 0 mg if there is no or minimal withdrawal (COWS < 5);
- 4 mg if there is mild withdrawal (COWS 5-12);
- 8 mg if there is moderate to severe withdrawal (COWS ≥ 13).

The suggested target total dose for Day One is in the range of 8 – 16 mg SUBOXONE FILM. A maximum daily dose of 32 mg should not be exceeded.

During the initiation of treatment, patients need frequent monitoring. SUBOXONE FILM should be dispensed in multiple doses over the first 4 to 6 hours of the transfer. Dosing supervision is recommended to ensure proper placement of the dose and to observe patient response to treatment as a guide to effective dose titration according to clinical effect.

<u>Dosage adjustment and maintenance</u>
The dose of SUBOXONE FILM should be increased progressively according to the clinical effect in the individual patient. The dose of SUBOXONE FILM should be adjusted progressively according to the clinical effect in the individual patient. The dosage is adjusted in increments or decrements of 2 – 8 mg buprenorphine to a level that maintains the patient in treatment and suppresses opioid withdrawal effects according to reassessments of the clinical and psychological status of the patient.

Most patients require daily buprenorphine doses in the range 12-24 mg to achieve stabilisation, although some patients require higher (e.g. up to 32 mg/day) or lower (4-8 mg/day) doses to achieve their treatment goals. During maintenance therapy, it may be necessary to periodically restabilise patients to new maintenance doses in response to changing patient needs.

During the initiation of treatment, daily dispensing of SUBOXONE FILM is recommended. After stabilisation, a reliable patient may be given a supply of SUBOXONE FILM sufficient for several days of treatment. It is recommended that the amount of SUBOXONE FILM be limited to 7 days according to local requirements.

Switching between SUBOXONE FILM strengths
The sizes and the compositions of units of SUBOXONE FILMS, i.e. 2 mg/0.5 mg and 8 mg/2 mg, are different from one another. If patients switch between various combinations of lower and higher strength units of SUBOXONE FILMS to obtain the same total dose (e.g. from four 2 mg/0.5 mg units to a single 8mg/2 mg unit, or vice-versa), systemic exposures of buprenorphine and naloxone may be different and patients should be monitored for over-dosing or under-dosing. For this reason, pharmacist should not substitute one or more film strengths for another without approval of the prescriber.

Switching between sublingual and buccal sites of administration
The systemic exposure of buprenorphine between buccal and sublingual administration of SUBOXONE FILM is similar. Therefore, once induction is complete, patients can switch between buccal and sublingual administration without significant risk of under or overdosing.

<u>Less than daily dosing of SUBOXONE FILM</u>
After a satisfactory period of stabilisation has been achieved the frequency of dosing may be decreased to dosing every other day at twice the individually titrated daily dose. For example, a patient stabilised to receive a daily dose of 8 mg may be given 16 mg on alternate days, with no medication on the intervening days. However, the dose given on any one day should not exceed 32 mg.

In some patients, after a satisfactory period of stabilisation has been achieved, the frequency of SUBOXONE FILM dosing may be decreased to 3 times a week (for example on Monday, Wednesday and Friday). The dose on Monday and Wednesday should be twice the individually titrated daily dose, and the dose on Friday should be three times the individually titrated daily dose, with no medication on the intervening days. However, the dose given on any one day should not exceed 32 mg.

The patient should be observed following the first multi-dose administration to initiate the less-than-daily dosing regimen, and whenever treated with high doses. Patients who sporadically use concomitant CNS-active medications or substances should be monitored closely.

Dosage reduction and termination of treatment
After a satisfactory stabilisation has been achieved, if the patient agrees, the dosage may be reduced gradually to a lower maintenance dose. In some favourable cases, treatment may be discontinued as part of a comprehensive treatment plan. The availability of SUBOXONE FILM in doses of 2 mg and 8 mg allows for a downward titration of dosage. Patients should be monitored following termination of treatment because of the potential for relapse.

Transferring between SUBOXONE sublingual tablet and SUBOXONE FILM
Patients being transferred between SUBOXONE sublingual tablet and SUBOXONE FILM should be started on the same dosage as the previously administered product. However, dosage adjustments may be necessary when switching between products. Due to the potentially greater relative bioavailability of SUBOXONE FILM compared to SUBOXONE sublingual tablet, patients switching from tablets to films should be monitored for over-medication. Those switching from SUBOXONE FILM to SUBOXONE sublingual tablet should be monitored for withdrawal or other indications of under-dosing. In clinical studies, pharmacokinetics of SUBOXONE FILMS was similar to the respective dosage strengths of SUBOXONE sublingual tablets; although not all doses and dose combinations met bioequivalence criteria. Furthermore, an open-label clinical study indicated that the dosage forms were clinically interchangeable.

Elderly
The safety and efficacy of buprenorphine in elderly patients over 65 years have not been established.

Pediatrics
SUBOXONE FILM is not recommended for use in children below age 15 years due to lack of data on safety and efficacy. Due to limited amount of available data, patients between 16 and 18 years of age should be closely monitored during treatment.

Dosage adjustment in hepatic impairment
Use of SUBOXONE FILM is contraindicated in patients with severe hepatic impairment.

SUBOXONE FILM may not be appropriate for patients with moderate hepatic impairment. SUBOXONE FILM may be used with caution for maintenance treatment in patients with moderate hepatic impairment.

Patients with moderate hepatic impairment prescribed SUBOXONE FILM should be monitored for signs and symptoms of precipitated opioid withdrawal. In addition, lower initial doses and cautious titration of dosage may be required in patients with moderate hepatic impairment.

No dosage adjustment is needed in patients with mild hepatic impairment.

Patients with impaired renal function
Renal elimination plays a relatively small role (~30%) in the overall clearance of buprenorphine. Therefore, no dose modification based on renal function is generally required. Metabolites of buprenorphine accumulate in patients with renal failure. Caution is recommended when dosing patients with severe renal impairment (CLcr <30 ml/min), which may require dose adjustment.

DRUG INTERACTIONS

SUBOXONE FILM should not be taken together with:

- Alcoholic drinks or medications containing alcohol, as alcohol increases the sedative effect of buprenorphine (see **Effects on ability to drive and use machines**).

SUBOXONE FILM should be used cautiously when co-administered with:

- Benzodiazepines: This combination may result in death due to respiratory depression of central origin; therefore, patients must be closely monitored when prescribed this combination, and this combination should be avoided where there is a risk of misuse. Patients should be warned that it is extremely dangerous to self-administer non-prescribed benzodiazepines while taking this product and should also be cautioned to use benzodiazepines concurrently with this product only as prescribed (see **PRECAUTIONS**).

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at $GABA_A$ sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see **PRECAUTIONS**).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

- Other central nervous system depressants, other opioid derivatives (e.g. methadone, analgesics and antitussives), certain antidepressants, sedative H1- receptor antagonists, barbiturates, anxiolytics other than benzodiazepines, neuroleptics, clonidine and related substances: these combinations increase central nervous system depression. The reduced level of alertness can make driving and using machines hazardous.
- Serotonergic Drugs: The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue SUBOXONE FILM if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (see **PRECAUTIONS**).
- Opioid analgesics: The analgesic properties of other opioids such as methadone and level III analgesics may be reduced in patients receiving treatment with buprenorphine/naloxone for opioid dependence. Adequate analgesia may be difficult to achieve when administering a full opioid agonist in patients receiving SUBOXONE FILM. Conversely, the potential for overdose should be considered with higher than usual doses of full agonist opioids, such as methadone or level III analgesics, especially when attempting to overcome buprenorphine partial agonist effects, or when buprenorphine plasma levels are declining. Patients with a need for analgesia and opioid dependence treatment may be best managed by multidisciplinary teams that include both pain and opioid dependence treatment specialists (see **PRECAUTIONS**).
- Naltrexone and other opioid antagonists: Since buprenorphine is a partial mu-opioid agonist, concomitantly administered opioid antagonists such as naltrexone can reduce or completely block the effects of SUBOXONE FILM. Patients maintained on SUBOXONE FILM may experience a sudden onset of prolonged and intense opioid withdrawal symptoms if dosed with opioid antagonists that achieve pharmacologically relevant systemic concentrations.
- CYP3A4 inhibitors: an interaction study of buprenorphine with ketoconazole (a potent inhibitor of CYP3A4) resulted in increased C_{max} and AUC (area under the curve) of buprenorphine (approximately 50% and 70% respectively) and, to a lesser extent, of norbuprenorphine. Patients receiving SUBOXONE FILM should be closely monitored and may require dose-reduction if combined with potent CYP3A4 inhibitors (e.g. protease inhibitors like

ritonavir, nelfinavir or indinavir or azole antifungals like ketoconazole or itraconazole, calcium channel antagonists, and macrolide antibiotics).

- CYP3A4 inducers: Concomitant use of CYP3A4 inducers with buprenorphine may decrease buprenorphine plasma concentrations, potentially resulting in under-treatment of opioid dependence with buprenorphine. Therefore it is recommended that patients receiving SUBOXONE FILM should be closely monitored if inducers (e.g. phenobarbital, carbamazepine, phenytoin and rifampicin) are co-administered.

ADVERSE EFFECTS

Safety Study of SUBOXONE FILM

The clinical safety of SUBOXONE FILM was evaluated in a trial (RB-US-07-0001) of 382 patients stabilised on SUBOXONE sublingual tablets for at least 30 days and then switched to SUBOXONE FILM for maintenance treatment. Two hundred and forty-nine (249) patients completed at least 12 weeks of dosing with the SUBOXONE FILM. Patients received SUBOXONE FILM sublingually or buccally in a 1:1 ratio (N=194 sublingually, N=188 buccally). Adjunctive treatment was “treatment as usual” with varying levels of counselling and behavioural treatment. Treatment was conducted on an outpatient basis. Among all patients who received SUBOXONE FILM either sublingually or buccally, the most common treatment emergent adverse events were oral mucosal erythema, sinusitis, nausea, toothache, pain and upper respiratory tract infection. The most common treatment emergent adverse event for the patients administered SUBOXONE FILM sublingually was upper respiratory tract infection (4 patients, 2.1%) and for patients administered SUBOXONE FILM buccally were oral mucosal erythema (6 patients, 3.2%), nausea (4 patients, 2.1%) and sinusitis (4 patients, 2.1%). All other adverse events were reported in 3 (1.5% or 1.6%, respectively) or fewer patients.

Adverse events reported to occur to at least 1% of patients being treated with SUBOXONE FILM in this trial are shown in Table 3.

Table 3 Adverse Events (≥1%) by Body System and Treatment Group in Study RB-US-07-0001, Sublingual and Buccal Administration

System Organ Class Preferred term	**Sublingual N=194**	**Buccal N=188**
Infections and Infestations		
Sinusitis	3 (1.5%)	4 (2.1%)
Upper respiratory tract infection	4 (2.1%)	2 (1.1%)
Pharyngitis streptococcal	2 (1.0%)	2 (1.1%)
Urinary tract infection	3 (1.5%)	1 (0.5%)
Influenza	2 (1.0%)	1 (0.5%)
Tooth abscess	2 (1.0%)	0 (0%)
Nasopharyngitis	0 (0%)	3 (1.6%)
Cellulitis	0 (0%)	2 (1.1%)
Gastrointestinal Disorders		
Glossodynia	3 (1.5%)	1 (0.5%)
Hypoaesthesia oral	2 (1.0%)	1 (0.5%)
Nausea	3 (1.5%)	3 (1.6%)
Oral mucosal erythema	2 (1.0%)	6 (3.2%)
Toothache	2 (1.0%)	4 (2.1%)
Vomiting	3 (1.5%)	2 (1.1%)

System Organ Class Preferred term	**Sublingual N=194**	**Buccal N=188**
Gastrooesophageal reflux disease	1 (0.5%)	3 (1.6%)
Constipation	1 (0.5%)	2 (1.1%)
Musculoskeletal and Connective Tissues Disorders		
Back pain	3 (1.5%)	1 (0.5%)
Arthralgia	2 (1.0%)	0 (0%)
Musculoskeletal pain	2 (1.0%)	0 (0%)
Muscle spasms	0 (0%)	2 (1.1%)
Psychiatric Disorders+		
Stress	2 (1.0%)	1 (0.5%)
Drug dependence (craving)	0 (0%)	2 (1.1%)
Injury, Poisoning and Procedural Complications		
Skin laceration	2 (1.0%)	1 (0.5%)
Road traffic accident	0 (0%)	2 (1.1%)
General Disorders and Administration Site Conditions		
Pain	3 (1.5%)	3 (1.6%)
Oedema peripheral	1 (0.5%)	2 (1.1%)
Nervous System Disorders		
Headache	2 (1.0%)	3 (1.6%)
Migraine	1 (0.5%)	2 (1.1%)
Renal and Urinary Disorders		
Nephrolithiasis	2 (1.0%)	2 (1.1%)
Metabolism and Nutrition Disorders		
Gout	1 (0.5%)	2 (1.1%)
Respiratory, Thoracic and Mediastinal Disorders		
Cough	0 (0%)	2 (1.1%)
Skin and Subcutaneous Tissue Disorders		
Dermatitis contact	2 (1.0%)	0 (0%)
Pregnancy, Puerperium and Perinatal Conditions		
Pregnancy	2 (1.0%)	0 (0%)

* AEs are coded using Medical Dictionary for Regulatory Activities (MedDRA) version 11.0 terminology.

Clinical trials of SUBOXONE sublingual tablets

Adverse events reported to occur to at least 1% of patients being treated in clinical trials of SUBOXONE sublingual tablets (CR96/013 + CR96/014) are shown in Tables 4 and 5.

Table 4 Adverse Events (≥1%) by Body System and Treatment Group in Study CR96/013

Body System/ Adverse Event (COSTART Terminology)	SUBOXONE (buprenorphine/naloxone) sublingual tablets 16/4 mg/day N=107 n (%)	SUBUTEX (buprenorphine) sublingual tablets 16 mg/day N=103 n (%)	Placebo N=107 n (%)	All Subjects (N = 317) n (%)
Body as a Whole				
Abscess	2 (1.9%)	1 (1.0%)	1 (0.9%)	4 (1.3%)
Asthenia	7 (6.5%)	5 (4.9%)	7 (6.5%)	19 (6.0%)
Chills	8 (7.5%)	8 (7.8%)	8 (7.5%)	24 (7.6%)
Fever	3 (2.8%)	3 (2.9%)	4 (3.7%)	10 (3.2%)
Headache	39 (36.4%)	30 (29.1%)	24 (22.4%)	93 (29.3%)
Infection	6 (5.6%)	12 (11.7%)	7 (6.5%)	25 (7.9%)
Accidental Injury	2 (1.9%)	5 (4.9%)	5 (4.7%)	12 (3.8%)
Pain	24 (22.4%)	19 (18.4%)	20 (18.7%)	63 (19.9%)
Pain abdomen	12 (11.2%)	12 (11.7%)	7 (6.5%)	31 (9.8%)
Pain back	4 (3.7%)	8 (7.8%)	12 (11.2%)	24 (7.6%)
Withdrawal syndrome	27 (25.2%)	19 (18.4%)	40 (37.4%)	86 (27.1%)
Cardiovascular System				
Vasodilation	10 (9.3%)	4 (3.9%)	7 (6.5%)	21 (6.6%)
Digestive System				
Constipation	13 (12.1%)	8 (7.8%)	3 (2.8%)	24 (7.6%)
Diarrhea	4 (3.7%)	5 (4.9%)	16 (15.0%)	25 (7.9%)
Dyspepsia	4 (3.7%)	5 (4.9%)	5 (4.7%)	14 (4.4%)
Nausea	16 (15.0%)	14 (13.6%)	12 (11.2%)	42 (13.2%)
Vomiting	8 (7.5%)	8 (7.8%)	5 (4.7%)	21 (6.6%)
Metabolic/Nutritional Disorders				
Peripheral Edema	1 (0.9%)	1 (1.0%)	2 (1.9%)	4 (1.3%)
Musculoskeletal System				
Myalgia	4 (3.7%)	1 (1.0%)	1 (0.9%)	6 (1.9%)
Nervous System				
Agitation	3 (2.8%)	2 (1.9%)	0	5 (1.6%)
Anxiety	3 (2.8%)	5 (4.9%)	4 (3.7%)	12 (3.8%)
Dizziness	5 (4.7%)	3 (2.9%)	4 (3.7%)	12 (3.8%)
Hyperkinesia	3 (2.8%)	2 (1.9%)	0	5 (1.6%)
Hypertonia	2 (1.9%)	0	2 (1.9%)	4 (1.3%)
Insomnia	15 (14.0%)	22 (21.4%)	17 (15.9%)	54 (17.0%)

Body System/ Adverse Event (COSTART Terminology)	SUBOXONE (buprenorphine/naloxone) sublingual tablets 16/4 mg/day N=107 n (%)	SUBUTEX (buprenorphine) sublingual tablets 16 mg/day N=103 n (%)	Placebo N=107 n (%)	All Subjects (N = 317) n (%)
Nervousness	5 (4.7%)	6 (5.8%)	4 (3.7%)	15 (4.7%)
Paresthesia	3 (2.8%)	3 (2.9%)	0	6 (1.9%)
Somnolence	8 (7.5%)	4 (3.9%)	2 (1.9%)	14 (4.4%)
Thinking Abnormal	2 (1.9%)	1 (1.0%)	1 (0.9%)	4 (1.3%)
Tremor	2 (1.9%)	1 (1.0%)	2 (1.9%)	5 (1.6%)
Respiratory System				
Cough Increased	1 (0.9%)	2 (1.9%)	2 (1.9%)	5 (1.6%)
Pharyngitis	2 (1.9%)	4 (3.9%)	1 (0.9%)	7 (2.2%)
Rhinitis	5 (4.7%)	10 (9.7%)	14 (13.1%)	29 (9.1%)
Skin And Appendages				
Sweating	15 (14.0%)	13 (12.6%)	11 (10.3%)	39 (12.3%)
Special Senses				
Amblyopia	3 (2.8%)	1 (1.0%)	0	4 (1.3%)
Lacrimation Disorder	0	4 (3.9%)	6 (5.6%)	10 (3.2%)
Urogenital System				
Dysmenorrhea	2 (1.9%)	1 (1.0%)	2 (1.9%)	5 (1.6%)
Urinary Tract Infection	1 (0.9%)	1 (1.0%)	2 (1.9%)	4 (1.3%)

Abbreviations: COSTART = Coding Symbols for Thesaurus of Adverse Reaction Terms.

Table 5 Adverse Events (≥1%) by Body System and Treatment Group in Study CR96/014

Body System/ Adverse Event (COSTART Terminology)	All SUBOXONE sublingual tablet Subjects N=472 n (%)
Body as a Whole	
Abscess	17 (3.6%)
Allergic Reaction	8 (1.7%)
Asthenia	48 (10.2%)
Chills	44 (9.3%)
Cyst	7 (1.5%)
Edema, Face	8 (1.7%)
Fever	36 (7.6%)
Flu Syndrome	89 (18.9%)
Headache	202 (42.8%)
Infection	5 (1.1%)
Infection, Viral	5 (1.1%)
Accidental Injury	72 (15.3%)
Malaise	9 (1.9%)
Neck Rigid	5 (1.1%)

Body System/ Adverse Event (COSTART Terminology)	All SUBOXONE sublingual tablet Subjects N=472 n (%)
Pain	197 (41.7%)
Pain, Abdomen	77 (16.3%)
Pain, Back	132 (28.0%)
Pain, Chest	23 (4.9%)
Pain, Neck	12 (2.5%)
Withdrawal Syndrome	194 (41.1%)
Cardiovascular System	
Hypertension	17 (3.6%)
Migraine	13 (2.8%)
Vasodilation	29 (6.1%)
Digestive System	
Abscess, Periodontal	10 (2.1%)
Anorexia	16 (3.4%)
Constipation	115 (24.4%)
Diarrhea	50 (10.6%)
Dyspepsia	45 (9.5%)
Flatulence	11 (2.3%)
Gastrointestinal Disorder	
Liver Function Abnormal	18 (3.8%)
Nausea	76 (16.1%)
Stomatitis	5 (1.1%)
Tooth Disorder	37 (7.8%)
Ulcer, Mouth	6 (1.3%)
Vomiting	61 (12.9%)
Hemic/Lymphatic System	
Anemia	7 (1.5%)
Ecchymosis	6 (1.3%)
Lymphadenopathy	5 (1.1%)
Metabolic/Nutritional Disorders	
Peripheral Edema	24 (5.1%)
Hyperglycemia	5 (1.1%)
Weight Decreased	15 (3.2%)
Musculoskeletal System	
Arthralgia	20 (4.2%)
Arthritis	5 (1.1%)
Leg Cramps	13 (2.8%)
Joint Disorder	9 (1.9%)
Myalgia	31 (6.6%)
Nervous System	
Agitation	10 (2.1%)
Anxiety	65 (13.8%)
Depression	70 (14.8%)
Dizziness	33 (7.0%)
Dream Abnormalities	9 (1.9%)
Drug Dependence	9 (1.9%)
Hypertonia	9 (1.9%)
Insomnia	138 (29.2%)
Libido Decreased	9 (1.9%)
Nervousness	42 (8.9%)

Body System/ Adverse Event (COSTART Terminology)	All SUBOXONE sublingual tablet Subjects N=472 n (%)
Paresthesia	28 (5.9%)
Somnolence	40 (8.5%)
Thinking Abnormal	6 (1.3%)
Tremor	7 (1.5%)
Respiratory System	
Asthma	21 (4.4%)
Bronchitis	9 (1.9%)
Cough Increased	36 (7.6%)
Dyspnea	9 (1.9%)
Lung Disorder	10 (2.1%)
Pharyngitis	64 (13.6%)
Pneumonia	12 (2.5%)
Respiratory Disorder	7 (1.5%)
Rhinitis	75 (15.9%)
Sinusitis	7 (1.5%)
Sputum Increased	5 (1.1%)
Yawn	6 (1.3%)
Skin and Appendages	
Acne	5 (1.1%)
Dermatological Contact	5 (1.1%)
Herpes Simplex	6 (1.3%)
Nodule, Skin	6 (1.3%)
Pruritus	11 (2.3%)
Skin Dry	6 (1.3%)
Sweat	74 (15.7%)
Urticaria	6 (1.3%)
Special Senses	
Amblyopia	5 (1.1%)
Conjunctivitis	14 (3.0%)
Eye Disorder	8 (1.7%)
Lacrimation Disorder	14 (3.0%)
Pain, Ear	8 (1.7%)
Urogenital System	
Dysmenorrhea	19 (4.0%)
Dysuria	9 (1.9%)
Hematuria	8 (1.7%)
Impotence	11 (2.3%)
Urinary Tract Infection	19 (4.0%)
Urine Abnormality	12 (2.5%)
Vaginitis	11 (2.3%)

The most common (≥10%) adverse events reported were those related to withdrawal symptoms (e.g. insomnia, headache, constipation, nausea, abdominal pain, diarrhoea, muscle aches, anxiety, sweating). In patients with marked opioid dependence, initial administration of buprenorphine can produce a withdrawal effect similar to that associated with naloxone.

Note - Patients enrolled in study RB-US-07-0001 on the soluble film were on a stable buprenorphine treatment prior to study initiation, while patients enrolled in studies CR96/013 and CR96/014 were buprenorphine-naïve individuals. As a result, the number of AEs observed in study RB-US-07-0001 is likely to be lower than that observed in studies CR96/013 and CR96/014.

As with other opioids, orthostatic hypotension can occur (see **PRECAUTIONS**).

Post marketing experience with opioids:
Serotonin syndrome (See **PRECAUTIONS**)

Adrenal insufficiency (See **PRECAUTIONS**)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

Post-marketing experience with buprenorphine alone and SUBOXONE
Post-marketing experience with buprenorphine alone has been associated with the following side effects: respiratory depression (see **PRECAUTIONS**) and coma, hallucinations, neonatal withdrawal syndrome, neonatal tremor, neonatal feeding disorder, foetal disorders, convulsions, confusion, miosis, weight decrease, asphyxia, hypoventilation, urinary retention, vertigo, drug dependence, headache, nausea, vomiting, drug withdrawal syndrome, peripheral oedema, heart rate and rhythm disorders, and deaths.

Cases of hepatitis with jaundice, hepatic failure, hepatic necrosis, hepatorenal syndrome, hepatic encephalopathy, and elevations in hepatic transaminases have been reported with buprenorphine use (see **PRECAUTIONS**).

In cases of intravenous misuse of buprenorphine, local reactions, sometimes septic potentially serious acute hepatitis, pneumonia, endocarditis and other serious infections have been reported.

Cases of acute or chronic hypersensitivity have been reported with buprenorphine with symptoms including rashes, hives, pruritus and reported cases of bronchospasm, angioneurotic oedema, and anaphylactic shock (see **PRECAUTIONS** and **CONTRAINDICATIONS**).

Very rare (<0.01%) side effects: loss of consciousness, cognitive disorders, psychosis, hallucinations, suicidal ideation, disorders of pregnancy (such as miscarriage and termination of pregnancy, premature birth, placental abruption, prolonged labour), foetal and neonatal complications (such as foetal disorders, foetal malformation, foetal growth retardation, foetal cystic hygroma, micrognathia, decreased oxygen saturation, developmental speech disorder, foetal dwarfism, foetal asphyxia, foetal cardiac rhythm disorder, cleft palate, Klinefelter's Syndrome, intersexual genitalia, neonatal withdrawal syndrome, neonatal tremor, neonatal feeding disorder, infant respiratory distress syndrome and subarachnoid bleeding), heart murmur, convulsions, confusion, miosis, weight decrease, asphyxia, hypoventilation, pruritus, angioedema, heart rate and rhythm disorders, pulmonary oedema, septic shock, infections (including sepsis, septic arthritis and septic embolus, staphylococcal sacroileitis, brain abscess, pneumonia and endocarditis and amniotic fluid infection) events associated with intravenous misuse (such as cutaneous ulceration, eschar, lividoid and necrotic lesions and penile and scrotal lesion), aphasia, aphonia, slurred speech, diplopia, facial palsy, ascites and lympodoema, pulmonary oedema, pulmonary artery thrombosis, pericardial effusion, shock, cerebrovascular accident, Popeye syndrome, intracranial haemorrhage, nephropathy, colic, denutrition splenic infarction, electrolyte imbalance (such as hyperkalaemia, hyponatraemia and hypoglycaemia), deaths (including death from suicide and sudden infant death syndrome) and unusual reactions. The actual incidence of all cases is extremely low and must be taken in consideration with the co-morbidities, life-style, environmental factors, and concomitant illicit and licit opioid use of the population under treatment.

A post-marketing study looking at injecting practices in Australia suggested that the combination of buprenorphine and naloxone is less commonly injected than buprenorphine alone. Additionally, post-marketing experience with SUBOXONE sublingual tablets for treatment of opioid dependence has been associated with the following side effects: anxiety, hyperhidrosis, syncope, insomnia, reduced feeling, anorexia (see also Tables 4 and 5 above), amnesia, convulsions, blood in vomit, fatigue, jaundice, swollen joints, miscarriage, shortness of breath, and suicide ideation. Treatment with SUBOXONE has been associated with orthostatic hypotension.

Additionally, post-marketing experience with SUBOXONE sublingual tablets for treatment of opioid dependence has been associated very rarely (<0.01%) with the following side effects: attempted suicide, disorders of pregnancy (such as premature birth), foetal and neonatal complications (such as foetal disorders, foetal malformation, foetal growth retardation, foetal cystic hygroma, micrognathia, macrocephaly, meconium staining and aspiration, decreased oxygen saturation, neonatal aspiration, asphyxia, developmental speech disorder, foetal dwarfism, foetal asphyxia, foetal cardiac rhythm disorder, low birth weight, Klinefelter’s Syndrome, mitochondrial disease, abnormal behaviour, developmental delay, developmental speech disorder intersexual genitalia, neonatal withdrawal syndrome, neonatal tremor, neonatal feeding disorder, subarachnoid bleeding and sudden infant death syndrome), pancreatitis, loss of consciousness, depression of consciousness, coordination disturbance, hallucinations, psychosis, mental disturbance and altered mental state, cerebral oedema, heart rate and rhythm disorders, septic shock, infections (including sepsis, pneumonia, chorioamniotitis and amniotic fluid infection) events associated with intravenous misuse (such as cellulitis), blurred vision, papilloedema, ascites and peripheral oedema, renal failure, adrenal insufficiency, electrolyte imbalance (such as hyperkalaemia, hypocalcaemia, hypomagnesaemia, hyponatraemia and hypoglycaemia) and deaths (including death from suicide and sudden infant death syndrome). The actual incidence of all cases is extremely low and must be taken in consideration with the co-morbidities, life-style, environmental factors, and concomitant illicit and licit opioid use of the population under treatment.

Post-marketing experience with SUBOXONE FILM for the treatment of opioid dependence has been most frequently associated with the following; adverse reactions appearing in at least 1% of reports by healthcare professionals are included in Table 6.

Table 6 Spontaneous adverse drug reactions collected through post-marketing surveillance reported by body system	
System Organ Class	**Preferred term**
Nervous system disorders	Headache
Gastrointestinal disorders	Glossitis Nausea Stomatitis Tongue disorder Vomiting
Skin and subcutaneous disorders	Rash
General disorders and administration site conditions	Drug ineffective Drug withdrawal syndrome Oedema peripheral

Reporting of suspected adverse reactions
Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the local reporting system.

CONTRAINDICATIONS
SUBOXONE FILM is contraindicated in the following instances:

- Hypersensitivity to buprenorphine or naloxone or any other component of the soluble film,
- Children less than 15 years of age,
- Severe respiratory or hepatic insufficiency (Child-Pugh C),
- Acute intoxication with alcohol or other CNS depressant.

PRECAUTIONS
Due to the limited amount of available data in adolescents (age 15 ≤ 18), SUBOXONE FILM should be used only with caution in this age group.

Patients should be closely monitored during the switching period from methadone to SUBOXONE FILM since withdrawal symptoms have been reported.

SUBOXONE FILM should be administered with caution in debilitated patients and those with impairment of hepatic, pulmonary, or renal function; myxoedema or hypothyroidism, adrenal cortical insufficiency (e.g. Addison's disease); CNS depression or coma; toxic psychoses; acute alcoholism; or delirium tremens.

Buprenorphine increases intracholedochal pressure as do other opioids. Therefore, caution should be exercised when SUBOXONE FILM is to be administered to patients with dysfunction of the biliary tract.

As with other opioids, caution is advised in patients using buprenorphine and having hypotension, prostatic hypertrophy or urethral stenosis.

<u>Misuse, abuse and diversion</u>
Diversion refers to the introduction of buprenorphine into the illicit market either by patients or by individuals who obtain the medicinal product through theft from patients or pharmacies. This diversion may lead to new addicts using buprenorphine as the primary opioid of abuse, with the risks of overdose, spread of blood borne viral infections, respiratory depression and hepatic injury. Because the naloxone in the combination film precipitated withdrawal in individuals dependent on heroin, methadone, or other full agonists, SUBOXONE FILM is expected to be less likely to be diverted for intravenous use. Sub-optimal treatment with SUBOXONE FILM may prompt medication misuse by the patient, leading to overdose or treatment dropout.

To minimise risk of misuse, abuse or diversion, appropriate precautions should be taken when prescribing and dispensing SUBOXONE FILM, such as to avoid prescribing multiple refills early in treatment, and to conduct patient follow-up visits with clinical monitoring that is appropriate to the patient's level of stability.

Patients dependent upon concomitant CNS-active substances, including alcohol, should not be treated with the increased doses required by the less-than-daily dosing regimen intended for use in a supervised dose setting. Patients with sporadic use of concomitant non-opioid medications should be monitored closely, and all patients dosed on a less-than-daily basis should be observed following the first multi-dose administration when initiating less-than-daily dosing or whenever treated with high doses.

<u>Precipitated withdrawal</u>
Because SUBOXONE FILM contains naloxone, it is highly likely to produce marked and intense opioid withdrawal symptoms if injected by patients treated with buprenorphine or SUBOXONE or by persons dependent on full opioid agonists such as heroin, oxycodone, morphine or methadone.

When initiating treatment with buprenorphine, the physician should be aware of the partial agonist profile of the molecule to the μ opioid receptors, which can precipitate withdrawal in opioid-dependent patients if given too soon after the administration of heroin, methadone or another opioid. To avoid precipitating withdrawal, induction with buprenorphine should be undertaken when objective and clear signs of withdrawal are evident.

Withdrawal symptoms may also be associated with suboptimal dosing.

A patient who is under-dosed with SUBOXONE may continue responding to uncontrolled withdrawal symptoms by self-medicating with opioids, alcohol or other sedative-hypnotics such as benzodiazepines.

<u>Dependence</u>
CAUTION: This preparation may be habit forming on prolonged use.
Buprenorphine is a partial agonist at the mu-opioid receptor and chronic administration produces dependence of the opioid type.

Discontinuation of treatment may result in a withdrawal syndrome that may be delayed. Studies in animals, as well as clinical experience, have shown that buprenorphine may produce dependence but at a lower level than morphine.

Consequently, it is important to follow the **DOSAGE AND ADMINISTRATION** recommendations.

<u>CNS depression</u>
SUBOXONE FILM may cause drowsiness, particularly when used together with alcohol or other central nervous system depressants (such as benzodiazepines, tranquilizers, sedatives or hypnotics) (see **DRUG INTERACTIONS**). When such combined therapy is contemplated, reduction of the dose of one or both agents should be considered.

SUBOXONE FILM should be used cautiously with MAOIs, based on experience with morphine.

<u>Respiratory depression</u>
SUBOXONE FILM is intended for sublingual or buccal use only. Significant respiratory depression has been associated with buprenorphine, particularly by the intravenous route. A number of cases of death due to respiratory depression have been reported, particularly when buprenorphine was used in combination with benzodiazepines (see **DRUG INTERACTIONS**), in opioid naïve individuals, or when buprenorphine was not used according to prescribing information.

Deaths have been reported in association with concomitant administration of buprenorphine and other depressants such as alcohol or other opioids.

Patients should be warned of the potential danger of the self-administration of benzodiazepines or other CNS depressants at the same time as receiving SUBOXONE FILM.

In the event of depression of respiratory or cardiac function, see **OVERDOSAGE INFORMATION**.

SUBOXONE FILM should be used with caution in patients with compromised respiratory function (e.g. chronic obstructive pulmonary disease, sleep apnoea, asthma, cor pulmonale, decreased respiratory reserve, hypoxia, hypercapnia, pre-existing respiratory depression or kyphoscoliosis).

SUBOXONE FILM may cause severe, possible fatal, respiratory depression in children who accidentally ingest it. Protect children against exposure.

Risks from Concomitant Use with Benzodiazepines
Profound sedation, respiratory depression, coma, and death may result from the concomitant use of SUBOXONE FILM with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when SUBOXONE FILM is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See **DRUG INTERACTIONS**).

Hepatitis, hepatic events
Cases of acute hepatic injury have been reported in opioid-dependent patients both in clinical trials and in post marketing adverse reaction reports. The spectrum of abnormalities ranges from transient asymptomatic elevations in hepatic transaminases to case reports of cytolytic hepatitis, hepatic failure, hepatic necrosis, hepatorenal syndrome, hepatic encephalopathy and death. Serious cases of acute hepatic injury have also been reported in a context of misuse, especially by the intravenous route. These hepatic injuries were dose-related and could be due to mitochondrial toxicity. Pre-existing or acquired mitochondrial impairment (genetic diseases, viral infections particularly chronic hepatitis C, liver enzyme abnormalities, alcohol abuse, anorexia, associated mitochondrial toxins, e.g. aspirin, isoniazid, valproate, amiodarone, antiviral nucleoside analogues, or drug misuse by injection) could promote the occurrence of such hepatic injuries. These co-factors must be taken into account before prescribing SUBOXONE FILM and during treatment monitoring. Baseline liver function tests and documentation of viral hepatitis status are recommended prior to commencing therapy. Patients who are positive for viral hepatitis, on concomitant medicines (see **DRUG INTERACTIONS**)

and/or have existing liver dysfunction are at greater risk of liver injury. Regular monitoring of liver function is recommended.

A further biological and etiological evaluation is recommended when a hepatic event is suspected. Depending upon the findings, the medicine may be discontinued cautiously so as to prevent withdrawal syndrome and to prevent a return to opioid dependence. If the treatment is continued, hepatic function should be monitored closely.

Hepatic impairment
Buprenorphine and naloxone are extensively metabolised by the liver. The effects of hepatic impairment on the pharmacokinetics of buprenorphine and naloxone were evaluated in a post-marketing study, in which a SUBOXONE 2 mg/0.5mg (buprenorphine/naloxone) sublingual tablet was administered to healthy subjects and subjects with varying degrees of hepatic impairment. Plasma levels were found to be elevated for buprenorphine and naloxone in patients with moderate to severe hepatic impairment (Table 7). Patients with severe hepatic impairment experienced substantially greater increases in exposure to naloxone relative to buprenorphine, and patients with moderate hepatic impairment experienced greater increases in exposure to naloxone relative to buprenorphine. The clinical impact in terms of efficacy/safety is unknown, but is likely to be greater for those with severe hepatic impairment than those with moderate hepatic impairment.

The doses of buprenorphine and naloxone in SUBOXONE FILM cannot be individually titrated. As such, SUBOXONE FILM should be avoided in patients with severe hepatic impairment. Use of SUBOXONE FILM may not be appropriate in those with moderate hepatic impairment. It may be used with caution for maintenance treatment in patients with moderate hepatic impairment who have initiated treatment on a buprenorphine-only product. Patients with moderate hepatic impairment should be monitored for signs and symptoms of precipitated opioid withdrawal. In addition, lower initial doses and cautious titration of dosage may be required in patients with moderate hepatic impairment. As with all patients treated with SUBOXONE FILM, liver function tests should be monitored prior to and during treatment. See also **DOSAGE AND ADMINISTRATION.**

Because buprenorphine is metabolised by the liver, its activity may be increased and/or extended in those individuals with impaired hepatic function. Naloxone metabolism may also be impaired in hepatic failure patients. The effects of hepatic impairment on the pharmacokinetics of buprenorphine and naloxone were evaluated in a post-marketing study in which a buprenorphine/naloxone 2 mg/0.5 mg sublingual tablet was administered to healthy subjects and subjects with varying degrees of hepatic impairment. Plasma levels for both buprenorphine and naloxone were found to be elevated in patients with moderate to severe hepatic impairment. Therefore, dose adjustments may be considered in cases of hepatic impairment, and patients should be monitored for signs and symptoms of precipitated opioid withdrawal.

Table 7 Effect of hepatic impairment on pharmacokinetic parameters of buprenorphine and naloxone following buprenorphine/naloxone administration (change relative to healthy subjects)

PK parameter	Mild Hepatic Impairment (Child-Pugh Class A) (n=9)	Moderate Hepatic Impairment (Child-Pugh Class B) (n=8)	Severe Hepatic Impairment (Child-Pugh Class C) (n=8)
BUPRENORPHINE			
C_{max}	1.2 fold increase	1.1 fold increase	1.7 fold increase
AUC_{last}	Similar to control	1.6 fold increase	2.8 fold increase
NALOXONE			
C_{max}	Similar to control	2.7 fold increase	11.3 fold increase
AUC_{last}	0.2 fold decrease	3.2 fold increase	14 fold increase

In the same study, changes in C_{max} and AUC_{last} of buprenorphine and naloxone in subjects with HCV infection without hepatic impairment were not clinically significant in comparison to the healthy subjects.

Effects on Laboratory Tests
Athletes should be aware that this medicine may cause a positive reaction to 'antidoping' tests.

Head Injury and Increased Intracranial Pressure
SUBOXONE FILM, like other potent opioids may itself elevate cerebrospinal fluid pressure, which may cause seizures and should be used with caution in patients with head injury, intracranial lesions and other circumstances where cerebrospinal pressure may be increased, or history of seizure.

SUBOXONE FILM can produce miosis and changes in the level of consciousness, or changes in the perception of pain as a symptom of disease and may interfere with patient evaluation or obscure the diagnosis or clinical course of concomitant disease.

As with other opioids, caution is advised in patients using buprenorphine and having hypotension, prostatic hypertrophy or urethral stenosis.

Opioids may produce orthostatic hypotension in ambulatory patients.

As with other mu-opioid receptor agonists, the administration of SUBOXONE FILM may obscure the diagnosis or clinical course of patients with acute abdominal conditions.

This product should be used with caution in patients with compromised respiratory function (e.g. chronic obstructive pulmonary disease, asthma, cor pulmonale, decreased respiratory reserve, hypoxia, hypercapnia, pre-existing respiratory depression or kyphoscoliosis.

Medicines that inhibit the enzyme CYP3A4 may give rise to increased concentrations of buprenorphine. A reduction of the SUBOXONE FILM dose may be needed. Patients already treated with CYP3A4 inhibitors should have their dose of SUBOXONE FILM titrated carefully since a reduced dose may be sufficient in these patients (see **DRUG INTERACTIONS**).

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs
Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of SUBOXONE FILM with serotonergic drugs (See **DRUG INTERACTIONS**). This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See **DRUG INTERACTIONS**). The onset of symptoms generally occurs within several hours to a few days of concomitant use but may occur later than that. Discontinue SUBOXONE FILM if serotonin syndrome is suspected.

Adrenal Insufficiency
Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence

of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual Function/Reproduction
Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Post marketing experience).

Effects on ability to drive and use machinery
SUBOXONE may influence the ability to drive and use machinery when administered to opioid dependent patients. This product may cause drowsiness, dizziness, or impaired thinking, especially during treatment induction and dose adjustment. If used together with alcohol or central nervous system depressants, the effect is likely to be more pronounced (see **PRECAUTIONS** and **INTERACTIONS**). Patients should be cautioned about operating hazardous machinery, including automobiles, until they are reasonably certain that SUBOXONE therapy does not adversely affect their ability to engage in such activities (see **DRUG INTERACTIONS** and **PRECAUTIONS**).

Use in Opioid Naïve Patients
There have been reported deaths of opioid naive individuals who received doses as low as 2 mg of buprenorphine sublingual tablet for analgesia. SUBOXONE is not appropriate as an analgesic.

PREGNANCY, LACTATION AND FERTILITY
Pregnancy
In rats, oral administration of buprenorphine at doses up to 20 times the maximum clinical dose of 32mg/day (based on mg/m^2) prior to and during gestation and lactation resulted in reduced implantation, fewer live births, and reduced pup weight gain and survival. There was no evidence of teratogenicity in rats and rabbits following parenteral administration of buprenorphine during the period of organogenesis, although there was embryofoetal toxicity, and reduced pup viability and developmental delays in rats. There was no evidence of teratogenicity in rats and rabbits following oral or intramuscular administration of maternally toxic doses of combinations of buprenorphine/naloxone during the period of organogenesis, although post-implantation losses were increased. In rats, oral (20 times maximum clinical dose, based on mg/m^2) or intramuscular administration of buprenorphine from late gestation to weaning was associated with increased stillbirths, reduced postnatal survival, and delayed postnatal development including weight gain and some neurological functions (surface righting reflex and startle response).

Buprenorphine readily crosses the placental barrier and may cause respiratory depression in neonates. During the last three months of pregnancy, chronic use of buprenorphine may be responsible for a withdrawal syndrome in neonates. SUBOXONE FILM should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus. Continued use of heroin during pregnancy is associated with significant risk to the mother and the foetus and neonate.

Data on the use of buprenorphine in pregnancy, and its impact on the mother and foetus, are limited. Data from randomised, controlled trials and observational studies do not indicate an increased risk of maternal or foetal adverse outcomes compared to methadone.

Chronic use of buprenorphine by the mother at the end of pregnancy may result in a withdrawal syndrome (e.g. hypertonia, neonatal tremor, neonatal agitation, myoclonus, apnoea, convulsions or bradycardia) in the neonate. In many reported cases the withdrawal was serious and required treatment. The syndrome is generally delayed for several hours to several days after birth (See **Pregnancy**). Due to the long half-life of buprenorphine, neonatal monitoring for several days should be considered to prevent the risk of respiratory depression or withdrawal syndrome in neonates.

Lactation
Animal studies indicate buprenorphine has the potential to inhibit lactation or milk production. In rats, oral (20 times maximum clinical dose, based on mg/m^2) or intramuscular administration of buprenorphine from late gestation to weaning was associated with increased stillbirths, reduced postnatal survival, and delayed postnatal development including weight gain and some neurological functions (surface righting reflex and startle response). The no effect level for developmental effects was twice the maximum clinical dose, based on mg/m^2. In two studies of thirteen women, buprenorphine was found in low levels in human breast milk. In both studies the estimated infant dose was <1% of the maternal dose. Because buprenorphine is excreted into human milk, the developmental and health benefits of breastfeeding should be considered along with the mother's clinical need for SUBOXONE FILM and any potential adverse effects on the breastfed child from the treatment or the underlying maternal condition.

Fertility
There were no effects on mating performance or fertility in rats following buprenorphine treatment at oral doses 20 times the maximum clinical dose of 32mg/day (based on mg/m^2). Dietary administration of SUBOXONE sublingual tablets to rats at doses of 47mg/kg/day or greater (estimated respective buprenorphine and naloxone exposures 14 and 24 times the anticipated clinical exposure, based on plasma AUC) resulted in reduced female conception rates. A dietary dose of 9.4mg/kg/day (twice the anticipated clinical exposure for both buprenorphine (based on AUC) and naloxone (based on mg/m^2) had no adverse effect on fertility.

OVERDOSAGE INFORMATION
Manifestations of acute overdose include miosis, sedation, hypotension, respiratory depression and death.

Nausea and vomiting may be observed.

The major symptom requiring intervention is respiratory depression, which could lead to respiratory arrest and death. If the patient vomits, care must be taken to prevent aspiration of the vomitus.

Treatment
In the event of depression of respiratory or cardiac function, primary attention should be given to the re-establishment of adequate respiratory exchange through provision of a patent airway and institution of assisted or controlled ventilation following standard intensive care measures. The patient should be transferred to an environment within which full resuscitation facilities are available.

Use of an opioid antagonist (i.e. naloxone) is recommended, despite the modest effect it may have in reversing the respiratory symptoms of buprenorphine compared with its effects on full agonist opioid agents.
Oxygen, intravenous fluids, vasopressors, and other supportive measures should be employed as indicated. High doses of naloxone hydrochloride 10-35 mg/70 kg may be of limited value in the management of buprenorphine overdose.

The long duration of action of SUBOXONE FILM should be taken into consideration when determining the length of treatment needed to reverse the effects of an overdose.

Naloxone can be cleared more rapidly than buprenorphine, allowing for a return of previously controlled buprenorphine overdose symptoms, so a continuing infusion may be necessary. Ongoing IV infusion rates should be titrated to patient response. If infusion is not possible, repeated dosing with naloxone may be required.

HOW SUPPLIED
Each soluble film is packed in an individual child resistant polyethylene terephthalate (PET)/low density polyethylene (LDPE)/aluminium/ethylene acrylic acid (EAA) sachet. There are 28 sachets in a pack.

STORAGE
The shelf-life of SUBOXONE FILM is 18 months. Store below 30°C. Keep out of reach of children.

For Medical Information and adverse event reporting please contact:
Indivior UK Limited
+800-270-81901
PatientSafetyRoW@indivior.com

Manufactured for:
Indivior UK Limited

Manufactured by:
Aquestive Therapeutics, Inc.
6560 Melton Road, Portage,
IN 46368, USA

Product Registration Holder:
Zuellig Pharma Sdn Bhd.
No.15, Persiaran Pasak Bumi, Seksyen U8,
Perindustrian Bukit Jelutong, 40150 Shah Alam,
Selangor Darul Ehsan

Date of Revision of Package Insert: August 2020